

3(a)	<i>any four from:</i> 1 gene(s) involved in production of carotene obtained ; 2 from, another species / bacteria / daffodil / maize / rice ; 3 ref. to use of named enzyme ; e.g. restriction enzyme / restriction endonuclease / (DNA) ligase 4 ref. to plasmid / vector / gene gun ; 5 ref. to recombinant DNA ; 6 ref. to insertion of gene into rice, cells / embryos / genome / DNA / chromosome ; 7 gene(s), expressed / transcribed, to produce enzyme(s) that make carotene ; 8 AVP ; e.g. extra detail such as promoter	4
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Question	Answer	Marks
3(b)	any four from: 1 33–34 grams of genetically modified rice needed (to meet the RDA of a child) ; 2 would help to prevent (childhood) blindness ; 3 would help to prevent death (of children) ; 4 (seeds) affordable / AW ; 5 (seeds) scientifically proven to be safe ; 6 (continuing to campaign would cause) waste of government money ; 7 AVP ; - e.g. benefits outweigh concerns - plant-based GM (no GM animal welfare concerns) - would reduce the need for supplementing diets	4

Question	Answer	Marks
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